Lecture 8. Estimation of Basic Quantities for Other Sampling Schemes (Part 2)

Double Censoring, Interval Censoring, Cohort Life Table

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1 Other Sampling Schemes

- Interval censoring: $X_i \in (L_i, R_i]$, never observed exactly.
- Double censoring: each observation is LC, RC, or exact.
- LT + interval censoring, or LT + double censoring.

Estimation here is hard: there is no simple closed-form estimator, and inference is not easy.

2 Turnbull's Iterative Method (1974) for Double Censoring

Pure left censoring is rare; samples with both left and right censoring are more common. Turnbull (1974) proposed a modified product-limit estimator with no closed form, computed iteratively.

- **Idea.** Start with an estimate of S that ignores the left-censored data; use it to compute the *expected* number of deaths at each observed time contributed by the left-censored observations; update S; repeat until it stabilizes (a self-consistency / EM-type algorithm).
- **Notation.** $0 = t_0 < t_1 < \cdots < t_m$ the distinct observed times; d_j = deaths at t_j (possibly 0); c_j = left-censored observations at t_j; r_j = right-censored observations at t_j.

Definition 8.1: Turnbull (1974) algorithm

Step 0. Initial estimate $S_0(t_j)$, e.g. the KM ignoring the left-censored observations.

Step $(K{+}1).1$**.** For each left-censoring time t_i and $j \le i$, estimate $p_{ij} = P(t_{j-1} < X \le t_j \mid$

$X \le t_i$) by

$$p_{ij} = \frac{S_K(t_{j-1}) - S_K(t_j)}{1 - S_K(t_i)}.$$

Step $(K+1).2$. Estimated events at t_j: $\hat{d}_j = d_j + \sum_{i=j}^{m} c_i\, p_{ij}$ (each left-censored observation is distributed over the times before it).

Step $(K+1).3$. Compute the usual KM from the pseudo right-censored data with $\hat{d}_j$ events and r_j right-censored observations at t_j, ignoring the left-censored data: $Y_j = \sum_{l=j}^{m}(\hat{d}_l + r_l)$.

Repeat until $S_{K+1}(t_j) \approx S_K(t_j)$ for all t_j.

Remark 8.2: Why p_{ij} has this form

A boy left censored at t_i is known to have $X \le t_i$. Given that, the chance his event fell in $(t_{j-1}, t_j]$ is $P(t_{j-1} < X \le t_j)/P(X \le t_i) = [S(t_{j-1}) - S(t_j)]/[1 - S(t_i)]$. Replacing each left-censored observation by these fractional events is the E-step; recomputing KM is the M-step.

Example 8.3: Age at first marijuana use (K&M Section 1.17)

California high-school boys: left censored = used marijuana but cannot recall the age of first use; right censored = never used. Initial KM ignoring the 12 left-censored boys (K&M Table 5.1):

t_i	10	11	12	13	14	15	16	17	18	> 18	Total
c_i	0	0	0	1	2	3	2	3	1	0	12
d_i	4	12	19	24	20	13	3	1	0	4	100
r_i	0	0	2	15	24	18	14	6	0	0	79
Y_i	179	175	163	142	103	59	28	11	4	4	
$S_0(t_i)$	.978	.911	.804	.669	.539	.420	.375	.341	.341	0	

Step 1 at $t_4 = 13$ (one left-censored boy, $1 - S_0(13) = 0.331$):

$$p_{41} = \tfrac{1.000-0.978}{0.331} = 0.067, \quad p_{42} = \tfrac{0.978-0.911}{0.331} = 0.202,$$

$$p_{43} = \tfrac{0.911-0.804}{0.331} = 0.320, \quad p_{44} = \tfrac{0.804-0.669}{0.331} = 0.410.$$

Doing this for every i with $c_i > 0$ (K&M Table 5.2; e.g. the column $i = 9$ reads $0.034, 0.102, 0.161, 0.206, 0.197, 0.180, 0.068, 0.052, 0.000$), Step 2 gives

$$\hat{d}_1 = 4 + 0.067(1) + 0.048(2) + 0.039(3) + 0.036(2) + 0.034(3) + 0.034(1) = 4.487.$$

Step 3 (K&M Table 5.3).

t_i	10	11	12	13	14	15	16	17	18	> 18
$\hat{d}_i$	4.487	13.461	21.313	26.963	22.437	14.714	3.417	1.207	0	4
Y_i	191.000	186.513	173.052	149.739	107.775	61.338	28.624	11.207	4	4
$S_1(t_i)$	.977	.906	.794	.651	.516	.392	.345	.308	.308	0

Iterate until convergence. Note $Y_1 = 191 = 179 + 12$: all left-censored boys are now accounted for as (fractional) events.

3 Turnbull's Iterative Method (1976) for Interval-Censored Data

Let $0 = \tau_0 < \tau_1 < \cdots < \tau_m$ be a grid containing all L_i, R_i. Define $\alpha_{ij} = 1$ if $(\tau_{j-1}, \tau_j] \subseteq (L_i, R_i]$ and 0 otherwise, i.e. whether the event of subject i could have occurred at τ_j.

Definition 8.4: Turnbull (1976) algorithm

Step 1. Probabilities of an event at τ_j: $p_j = S(\tau_{j-1}) - S(\tau_j)$, $j = 1, \ldots, m$.

Step 2. Expected number of events at τ_j: $d_j = \sum_{i=1}^{n} \frac{\alpha_{ij} p_j}{\sum_k \alpha_{ik} p_k}$ (each subject's unit mass is split over its admissible grid points in proportion to p_j).

Step 3. Estimated number at risk: $Y_j = \sum_{k=j}^{m} d_k$.

Step 4. Update the product-limit estimator from these pseudo-data; stop if it is close to the previous S at all τ_j, else repeat Steps 1–3.

Example 8.5: Time to cosmetic deterioration, early breast cancer (radiotherapy only)

46 patients; data are intervals $(L, R]$ in months or right-censored "$\geq$" values: $(0, 7]$; $(0, 8]$; $(0, 5]$; $(4, 11]$; $(5, 12]$; $(5, 11]$; $(6, 10]$; $(7, 16]$; $(7, 14]$; $(11, 15]$; $(11, 18]$; ≥ 15; ≥ 17; $(17, 25]$; $(17, 25]$; ≥ 18; $(19, 35]$; $(18, 26]$; ≥ 22; ≥ 24; ≥ 24; $(25, 37]$; $(26, 40]$; $(27, 34]$; ≥ 32; ≥ 33; ≥ 34; $(36, 44]$; $(36, 48]$; ≥ 36; ≥ 36; $(37, 44]$; ≥ 37; ≥ 37; ≥ 37; ≥ 38; ≥ 40; ≥ 45; ≥ 46 ($\times 7$).

- **Initial estimate**: distribute each subject's mass $1/46$ equally over the grid points in $(L_i, R_i]$; e.g. the subject with $(0, 7]$ gives $(1/46)(1/4)$ to each of $\tau = 4, 5, 6, 7$.
- With S_0: $p_1 = 1 - 0.979 = 0.021$, $p_2 = 0.979 - 0.955 = 0.024$, etc.
- At $\tau = 4$, three intervals contain it ($(0, 8]$, $(0, 7]$, $(0, 5]$), so

$$d_1 = \frac{0.021}{0.021 + 0.024 + 0.021 + 0.029 + 0.031} + \frac{0.021}{0.021 + 0.024 + 0.021 + 0.029} + \frac{0.021}{0.021 + 0.024} = 0.842.$$

- One iteration changes S only slightly at early times (e.g. $0.979 \to 0.982$ at $\tau = 4$) but more in the tail (e.g. $0.229 \to 0.341$ at $\tau = 46$); iterate to convergence.

4 The Cohort Life Table

4.1 Setting

A **cohort** is a group of individuals with a common origin from which event times are measured. Each event or censoring time is recorded only as falling in one of $k + 1$ adjacent, non-overlapping intervals $(a_{j-1}, a_j]$, $j = 1, \ldots, k + 1$. A traditional cohort life table presents the actual mortality experience of the cohort from the birth of each individual to the death of the last surviving member; it summarizes time-to-event information over these intervals (grouped data).

Example 8.6: NLSY weaning data

National Labor Survey of Youth (K&M Section 1.14): 927 first-born infants who were breast fed and have complete information. Response: duration of breast feeding in weeks, with an indicator of whether the baby was weaned (event) or lost to follow-up (censored). First rows: weeks 0–2: 927 entering, 2 withdrawn; 2–3: 848 entering, 3 withdrawn; 3–5: 774 entering, 6 withdrawn; ...

4.2 Construction

Definition 8.7: Life-table quantities

1. Intervals $I_j = (a_{j-1}, a_j]$, $j = 1, \ldots, k+1$, $a_0 = 0$, $a_{k+1} = \infty$; each event/censoring time falls in exactly one of them.
2. Y'_j: number entering I_j without having experienced the event.
3. W_j: number lost to follow-up or withdrawn alive in I_j.
4. $Y_j = Y'_j - W_j/2$: estimated number at risk in I_j, assuming censoring times are uniform over the interval (each withdrawal is at risk for half the interval on average) — the *actuarial* adjustment.
5. d_j: number of events in I_j.
6. Survival at the start of I_j ($\hat{S}(a_0) = 1$):

$$\hat{S}(a_j) = \hat{S}(a_{j-1})[1 - d_j/Y_j] = \prod_{i=1}^{j} [1 - d_i/Y_i]. \tag{8.1}$$

7. Density at the midpoint $a_{mj} = (a_{j-1} + a_j)/2$:

$$\hat{f}(a_{mj}) = \frac{\hat{S}(a_{j-1}) - \hat{S}(a_j)}{a_j - a_{j-1}}. \tag{8.2}$$

8. Hazard at the midpoint (with $\hat{S}(a_{mj})$ taken as the average of the endpoint values):

$$\hat{h}(a_{mj}) = \frac{\hat{f}(a_{mj})}{\hat{S}(a_{mj})} = \frac{2\hat{f}(a_{mj})}{\hat{S}(a_j) + \hat{S}(a_{j-1})} = \frac{d_j}{(a_j - a_{j-1})(Y_j - d_j/2)}. \tag{8.3}$$

Derivation: deriving (8.3)

Write $\hat{q}_j = d_j/Y_j$, $\hat{p}_j = 1 - \hat{q}_j$ and $\Delta_j = a_j - a_{j-1}$. Then $\hat{S}(a_j) = \hat{S}(a_{j-1})\hat{p}_j$ and $\hat{f}(a_{mj}) = \hat{S}(a_{j-1})\hat{q}_j/\Delta_j$, so

$$\hat{h}(a_{mj}) = \frac{2\hat{S}(a_{j-1})\hat{q}_j/\Delta_j}{\hat{S}(a_{j-1})(1 + \hat{p}_j)} = \frac{2\hat{q}_j}{\Delta_j(1 + \hat{p}_j)} = \frac{2d_j/Y_j}{\Delta_j(2 - d_j/Y_j)} = \frac{d_j}{\Delta_j(Y_j - d_j/2)}.$$

Interpretation: events per unit of person-time, where those who fail in I_j are credited with half the interval.

4.3 Standard errors

Theorem 8.8: Estimated standard deviations

$$\widehat{\mathrm{SD}}[\hat{S}(a_{j-1})] = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1}\frac{d_i}{Y_i(Y_i-d_i)}} = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1}\frac{\hat{q}_i}{Y_i\hat{p}_i}}, \tag{8.4}$$

$$\widehat{\mathrm{SD}}[\hat{f}(a_{mj})] = \frac{\hat{S}(a_{j-1})\hat{q}_j}{a_j-a_{j-1}}\sqrt{\sum_{i=1}^{j-1}\frac{\hat{q}_i}{Y_i\hat{p}_i}+\frac{\hat{p}_j}{Y_j\hat{q}_j}}, \tag{8.5}$$

$$\widehat{\mathrm{SD}}[\hat{h}(a_{mj})] = \left\{\frac{1-[\hat{h}(a_{mj})(a_j-a_{j-1})/2]^2}{Y_j\hat{q}_j}\right\}^{1/2}\hat{h}(a_{mj}). \tag{8.6}$$

(8.4) is Greenwood's formula; the SD of $\hat{S}(a_0)=1$ is 0.

4.4 Other summary quantities

1. **Mean**: $\hat{\mu}=\int_0^\infty \hat{S}(t)\,\mathrm{d}t$. If the largest observation is censored, $\hat{S}$ stays flat and the tail must be completed in one of the three ways of Lecture 5 (Efron, Gill, Brown–Hollander–Kowar).
2. **Median** by linear interpolation: find I_j with $\hat{S}(a_{j-1})\ge 0.5\ge\hat{S}(a_j)$, then
$$\hat{x}_{0.5}=a_{j-1}+\frac{[\hat{S}(a_{j-1})-0.5](a_j-a_{j-1})}{\hat{S}(a_{j-1})-\hat{S}(a_j)}=a_{j-1}+\frac{\hat{S}(a_{j-1})-0.5}{\hat{f}(a_{mj})}. \tag{8.7}$$
3. **Mean residual life** at the start of I_i:
$$\widehat{\mathrm{mr}}(a_{i-1})=\frac{\int_{a_{i-1}}^\infty \hat{S}(t)\,\mathrm{d}t}{\hat{S}(a_{i-1})}=\frac{\sum_{j=i}^k \hat{S}(a_{j-1})(a_j-a_{j-1})}{\hat{S}(a_{i-1})}.$$
4. **Median residual life**: $\mathrm{mdrl}(a_{i-1})$ solves $S(a_{i-1})/2=S(a_{i-1}+\mathrm{mdrl}(a_{i-1}))$ — the time increment beyond which half of those alive at a_{i-1} are expected to survive.

Example 8.9: life table for weaning (K&M Table 5.6)

Weeks	Y'_j	W_j	Y_j	d_j	$\hat{S}(a_{j-1})$	$\hat{f}(a_{mj})$	$\hat{h}(a_{mj})$	SD $\hat{S}$	SD $\hat{f}$	SD $\hat{h}$
0–2	927	2	926	77	1.0000	0.0416	0.0434	0	0.0045	0.0049
2–3	848	3	846.5	71	0.9168	0.0769	0.0875	0.0091	0.0088	0.0104
3–5	774	6	771	119	0.8399	0.0648	0.0836	0.0121	0.0055	0.0076
5–7	649	9	644.5	75	0.7103	0.0413	0.0618	0.0149	0.0046	0.0071
7–11	565	7	561.5	109	0.6276	0.0305	0.0537	0.0160	0.0027	0.0051
11–17	449	5	446.5	148	0.5058	0.0279	0.0662	0.0166	0.0021	0.0053
17–25	296	3	294.5	107	0.3381	0.0154	0.0555	0.0158	0.0014	0.0052
25–37	186	0	186	74	0.2153	0.0071	0.0414	0.0138	0.0008	0.0047
37–53	112	0	112	85	0.1296	0.0061	0.0764	0.0114	0.0006	0.0066
53–	27	0	27	27	0.0313	—	—	0.0059	—	—

Check of row 1: $Y_1=927-2/2=926$; $\hat{q}_1=77/926=0.0832$, so $\hat{S}(2)=0.9168$; $\hat{f}=(1-0.9168)/2=0.0416$; $\hat{h}=77/\{2(926-38.5)\}=0.0434$.

Median: $\hat{S}(11)=0.5058\ge 0.5\ge\hat{S}(17)=0.3381$, so $\hat{x}_{0.5}=11+\dfrac{(0.5058-0.5)(17-11)}{0.5058-0.3381}\approx$ 11.2 weeks.

Hazard (K&M Fig. 5.3): high (about 0.08–0.09 per week) in the first 5 weeks, about 0.05–0.07 through week 25, lowest (about 0.04) in weeks 25–37, then rising again in weeks 37–53.